**Project Design Phase-I**
**Solution Architecture**

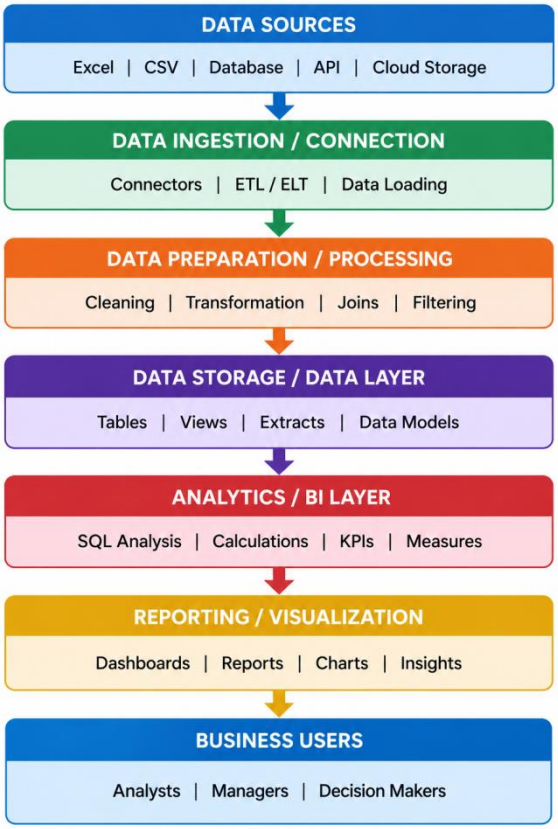
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|---|---|
| Date | 28 October 2026 |
| Team ID | EN25CS3010655 |
| Project Name | Health360: Comprehensive Healthcare Risk & Disease Analytics |
| Maximum Marks | 4 Marks |

## Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

## Sample Solution Architecture Diagram:



**Component Description Table**

| Component Name | Description / Role in Architecture | Technology Used |
|---|---|---|
| Data Sources | Collects raw physiological metrics, EHR records, and external data feeds (Excel, CSV, Database, API, Cloud Storage). | IoT Wearables, Medical Sensors, APIs, Cloud Storage |
| Data Ingestion / Connection | Handles secure data streaming, connectors, ETL/ELT workflows, and data loading. | MQTT, Apache Kafka, REST APIs |
| Data Preparation / Processing | Cleans, transforms, joins, and filters incoming vital signs and patient data. | Python (Pandas, NumPy), Apache Spark |
| Data Storage / Data Layer | Organizes and stores structured and unstructured records using tables, views, extracts, and data models. | PostgreSQL, MongoDB, AWS S3 |
| Analytics / BI Layer | Performs SQL analysis, calculations, health metrics tracking, KPIs, and machine learning measures. | Python (Scikit-Learn/TensorFlow), SQL |
| Reporting / Visualization | Delivers real-time dashboards, medical reports, charts, and actionable clinical insights. | React, Chart.js, HTML5/CSS |
| Business Users | End users including doctors, hospital managers, administrators, and patients who utilize the insights. | Web & Mobile Dashboards |